Rethinking the policy optimization with Gradient Estimators

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Abstract

Permissive policy optimization can learn rapidly from rollout data and then fail abruptly after many successful updates. Policy-deviation bounds explain why a large change can be unsafe, but they do not say when a finite rollout stops supporting an accurate update for the current policy. We study this transition through two ordered questions. First, when is the ordinary full-trajectory importance-sampling (IS) estimator still reliable? We show that normalized effective sample size (ESS) controls the policy-mismatch component of its finite-sample error and, through smoothness, yields a high-probability one-step improvement guarantee. Second, when does coefficient control provide a better update? Lower gradient-estimation MSE is not enough: the controlled estimator must also retain sufficient alignment with the population gradient. We capture this requirement with a full fixed-step improvement certificate. An exactly solvable Gaussian bandit shows that the two criteria can switch at different ESS levels: PPO can have lower MSE while IS still gives the larger exact expected improvement. Along the example's monotone off-policy path, the safe certificate oracle—IS, PPO, or no update—reduces exactly to two ESS thresholds. In the reported large-model trajectories, ESS-conditioned safeguards preserve early learning while preventing late failure.

1 Introduction

Reinforcement learning with verifiable rewards repeatedly collects responses from a rollout policy and reuses each batch for one or more policy updates. A permissive update can work surprisingly well in this setting: performance rises quickly and remains stable, but then collapses abruptly. Figure 1 shows this delayed failure. Rollout data support useful learning for many updates before the same update rule becomes unstable.

Classical theory explains why moving too far from the rollout policy can be dangerous [11, 13, 14]. That explanation is essential, but it does not explain why failure is delayed. A policy-deviation bound controls a population approximation; it does not track how much information a finite rollout still provides for the gradient of the current policy. The practical question raised by Figure 1 is therefore statistical: when does the rollout cease to support a reliable next update?

Our starting point is that nominal batch size can remain fixed while effective support contracts. Importance weights become increasingly uneven as the current policy moves away from the rollout policy, so fewer responses determine the update. Normalized ESS measures this contraction. We connect it first to gradient reliability and then to policy improvement.

This separation leads to two ordered questions. First, when is the IS importance-weighted gradient reliable enough to improve the population objective? Second, if IS becomes unreliable, when is a controlled estimator actually better? The second question cannot be answered by ESS or MSE alone. Clipping may reduce variance and MSE while also removing useful population-gradient alignment. We therefore compare candidate estimators using the full fixed-step improvement certificate.

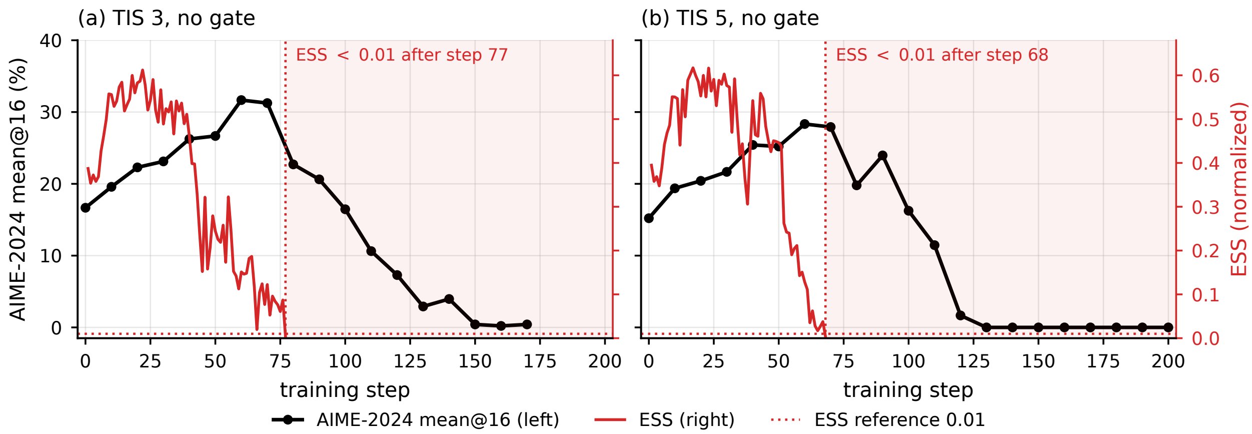

Figure 1: A delayed failure in Qwen3-30B-A3B training. The permissive runs improve for many updates and fail only after the effective support of the rollout data has sharply deteriorated. This transition is the phenomenon studied in the paper.

An exactly solvable Gaussian bandit makes the distinction concrete. Every population moment and the expected one-step gain are available without Monte Carlo error, allowing us to separate the MSE crossover from the actual improvement crossover. We then test the resulting regime prediction in large-model training. These experiments use ESS as a diagnostic of effective support and test the predicted support-to-intervention ordering.

1.1 Contributions

1. We show how normalized ESS controls the finite-sample error of the IS gradient estimator from a fixed rollout and yields a one-step population improvement guarantee.

2. We distinguish estimator accuracy from update quality and derive the fixed-step condition under which coefficient control has the stronger improvement certificate.

3. We separate the two crossovers exactly in a Gaussian bandit and then evaluate ESS-conditioned safeguards in large-model training.

2 Related work

Policy-deviation guarantees. Trust-region and proximal methods control policy movement to obtain stable improvement guarantees or practical update rules [11, 13, 14]. These analyses provide the standard explanation for why large policy changes can be unsafe. Our framework is complementary: it studies when a fixed finite rollout ceases to provide a reliable estimate of the current policy gradient. The distinction is between a population-level approximation guarantee and a finite-sample reliability question.

Importance sampling and effective support. Importance-sampling policy optimization uses likelihood-ratio moments and Rényi divergence to quantify the reliability of off-policy estimators [5, 6]. In these results, effective rather than nominal sample size controls estimation error. We apply the same principle directly to the vector-valued policy-gradient estimator and use normalized ESS as the regime coordinate.

Reliable policy-gradient updates. Smoothness-based analyses translate the error of a finite gradient estimate into a lower bound on policy improvement and use this relation to choose step sizes or sample sizes [8, 10]. We combine this optimization argument with an ESS-dependent off-policy reliability bound.

Stability in RLVR. Recent methods alter ratio boundaries, sequence weights, divergence controls, or update rules to improve the stability of large-model reinforcement learning [2, 3, 7, 9, 16, 18, 19]. Rather than proposing another static boundary, we study the statistical transition that makes a permissive update unreliable and then ask when an intervention can recover reliability.

3 Preliminaries

This section provides the background and notation used in the following analysis.

3.1 Policy-gradient estimators from a fixed rollout

Problem formulation. Let $X \sim \nu$ denote a context, $Q(\cdot \mid X)$ the rollout policy, and $P_\theta(\cdot \mid X)$ the current policy. A complete sample is $Z = (X, Y)$, where $Y = (Y_1, \dots, Y_T)$ and $R(Z)$ is its reward. For each factor t, define the likelihood ratio and score

$$r_{\theta,t}(Z) := \frac{P_\theta(Y_t \mid X, Y_{<t})}{Q(Y_t \mid X, Y_{<t})}, \qquad \ell_{\theta,t}(Z) := \nabla_\theta \log P_\theta(Y_t \mid X, Y_{<t}). \tag{1}$$

Here t indexes a policy factor, such as a token or decision step. Because both policies factorize along the trajectory, the complete-sample likelihood ratio is

$$W_\theta(Z) := \frac{P_\theta(Y \mid X)}{Q(Y \mid X)} = \prod_{t=1}^{T} r_{\theta,t}(Z). \tag{2}$$

The learning objective is to maximize the expected reward $J(\theta) := \mathbb{E}_{P_\theta}[R]$. Let $A(Z) := R(Z) - b(X)$ be a baseline-centered reward, where $b(X)$ does not depend on Y. Under $P_\theta(\cdot \mid X) \ll Q(\cdot \mid X)$, change of measure expresses the population gradient under the rollout distribution:

$$g(\theta) := \nabla J(\theta) = \mathbb{E}_Q[W_\theta f_\theta], \qquad f_\theta(Z) := A(Z) \sum_{t=1}^{T} \ell_{\theta,t}(Z) = A(Z) \nabla_\theta \log P_\theta(Y \mid X). \tag{3}$$

While trust-region arguments motivate safeguards such as TRPO and PPO-style clipping [13, 14], these arguments are typically made at the population level. Practical algorithms, however, do not observe $g(\theta)$ and must rely on its finite-sample counterpart. Given i.i.d. rollout samples $Z_i \sim \nu(dx)Q(dy \mid x)$, $i = 1, \dots, N$, the ordinary full-trajectory importance-sampling (IS) policy-gradient estimator is

$$\widehat{g}_N(\theta) := \frac{1}{N} \sum_{i=1}^{N} W_{\theta,i} f_{\theta,i}. \tag{4}$$

The importance-weighted nature of this estimator can produce poor agreement between the finite-sample update and the population gradient [5, 6]. This exposes an important but often overlooked question: when is the finite-sample gradient reliable?

Choice of gradient estimator. To formalize this comparison, write $\mathbf{r}_\theta = (r_{\theta,1}, \ldots, r_{\theta,T})$ and let $u = (u_1, \ldots, u_T)$ be a collection of coefficient rules, with $u_t : \mathbb{R}_+^T \times \mathbb{R} \to \mathbb{R}$. Define

$$\psi_\theta(u; Z) := A(Z) \sum_{t=1}^{T} u_t(\mathbf{r}_\theta, A(Z))\ell_{\theta,t}(Z), \qquad \widehat{g}_N(u; \theta) := \frac{1}{N} \sum_{i=1}^{N} \psi_\theta(u; Z_i). \tag{5}$$

Existing ratio-based policy-gradient estimators can be represented by particular choices of u. The IS estimator sets $u_{\mathrm{IS},t} = W_\theta$ for every factor. One popular alternative is PPO/GRPO ratio clipping, whose clipped surrogate induces the following gradient-masking rule with radius $0 < \epsilon < 1$ [14, 15]:

$$u_{\mathrm{P},t}(\mathbf{r}, A) := r_t \left[\mathbf{1}\{A \geq 0,\ r_t \leq 1 + \epsilon\} + \mathbf{1}\{A < 0,\ r_t \geq 1 - \epsilon\}\right]. \tag{6}$$

Such clipping rules create a conventional bias–variance tradeoff. More aggressive clipping can reduce sampling variation, but it can also increase bias by removing gradient terms. This raises the natural question: what governs the reliability of a gradient estimator, and when is clipping worth the signal it removes? We answer the first part by introducing Rényi divergence and effective sample size (ESS).

3.2 Gradient estimators and Rényi divergence

The dispersion of the importance weights indicates how dissimilar P_θ and Q are. Rényi divergence formalizes this notion as an information-theoretic dissimilarity between probability measures [12, 17], while importance-sampling analyses connect it to the finite-sample behavior of importance-weighted means [5, 6].

Rényi divergence is a family of measures indexed by a moment order $\alpha > 1$. For the current and rollout policies, define

$$D_\alpha(P_\theta \| Q) := \frac{1}{\alpha - 1} \log \mathbb{E}_Q[W_\theta^\alpha], \qquad d_\alpha(P_\theta \| Q) := \exp\{D_\alpha(P_\theta \| Q)\}. \tag{7}$$

Larger values of α place progressively more emphasis on rare, large likelihood ratios. The order-two case is central because $\mathbb{E}_Q[W_\theta^2]$ controls standard MSE bounds for bounded importance-weighted contributions. Importantly, its reciprocal is the normalized effective sample size [5, 6]:

$$\rho_\theta := \frac{1}{d_2(P_\theta \| Q)} = \frac{1}{\mathbb{E}_Q[W_\theta^2]} = \exp\{-D_2(P_\theta \| Q)\}. \tag{8}$$

When $\mathbb{E}_Q[W_\theta^2] < \infty$, Jensen's inequality and $\mathbb{E}_Q[W_\theta] = 1$ give $\rho_\theta \in (0, 1]$, and $N\rho_\theta$ is the effective sample count corresponding to nominal batch size N. As the second moment of the likelihood ratio grows, both ρ_θ and this effective count contract. The identity $\mathbb{E}_Q[(W_\theta - 1)^2] = \rho_\theta^{-1} - 1$ shows that ρ_θ is exactly the inverse second-moment measure of ratio dispersion.

Importance-sampling theory makes this quantity operational. If $\|f_\theta\|_\infty := \sup_z \|f_\theta(z)\|_2 < \infty$, the MSE of the IS estimator is at most $\|f_\theta\|_\infty^2/(N\rho_\theta)$. Normalized ESS therefore captures the policy-mismatch component of IS reliability. The next section determines when that reliability is sufficient for improvement and then asks whether coefficient control offers a better update.

4 Theoretical Analysis

This section answers the preceding questions in order. We first connect gradient-estimator reliability to policy improvement, then use normalized ESS to control the reliability of the IS estimator, and finally ask when clipping provides the stronger update.

4.1 Gradient-estimator reliability and policy improvement

The preceding section distinguishes the finite-sample IS estimate from the population gradient it targets. Here, we connect estimator reliability to a lower bound on policy improvement. The following lemma formalizes this connection for an arbitrary gradient estimator.

Lemma 1 (A reliable gradient estimate yields policy improvement)**.** *Let $\widehat{g}$ be an estimate of $g = \nabla J(\theta)$ satisfying $\|\widehat{g} - g\|_2 \leq \varepsilon$. Suppose J is L-smooth along the update segment for $L > 0$, and define $\gamma = L^{-1}(1 - \varepsilon/\|\widehat{g}\|_2)_+$, with $\gamma = 0$ when $\widehat{g} = 0$. Then*

$$J(\theta + \gamma\widehat{g}) - J(\theta) \geq \frac{1}{2L}\left(\|\widehat{g}\|_2 - \varepsilon\right)_+^2. \tag{9}$$

This deterministic lemma uses the same smoothness mechanism as existing policy-gradient analyses [8, 10]. Its implication is direct: the error radius measures how far the finite-sample direction can lie from the population gradient. When this radius is smaller than $\|\widehat{g}\|_2$, the estimator yields a positive lower bound on policy improvement.

ESS and the IS update. However, the population gradient is unknown, so this reliability cannot be evaluated directly. Normalized ESS provides the bridge: for the IS estimator, it converts likelihood-ratio dispersion into a bound on gradient-estimation error. At update k, condition on the preceding updates and write $W_k := W_{\theta_k}$, $f_k := f_{\theta_k}$, $g_k := g(\theta_k)$, $\widehat{g}_k := \widehat{g}_N(\theta_k)$, and $\rho_k := \rho_{\theta_k}$. Assume that the N rollout samples are independent draws from Q and that $\|f_k\|_\infty := \sup_z \|f_k(z)\|_2 < \infty$.

Because the IS estimator is unbiased, its exact mean-squared error is

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\left\{\mathbb{E}_Q[W_k^2\|f_k\|_2^2] - \|g_k\|_2^2\right\}. \tag{10}$$

Using $\|f_k(z)\|_2 \leq \|f_k\|_\infty$ and $\rho_k^{-1} = \mathbb{E}_Q[W_k^2]$ gives the interpretable upper bound

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 \leq \frac{\|f_k\|_\infty^2}{N\rho_k}. \tag{11}$$

Likelihood-ratio dispersion therefore enters through the effective count $N\rho_k$, replacing the nominal sample size N. Equation (11) controls the mean-squared error; the next lemma converts this result into a high-probability error radius for the realized update.

Lemma 2 (Normalized ESS controls the gradient-error radius)**.** *Under the preceding setup, suppose $d_2(P_{\theta_k}\|Q) < \infty$ and $\|f_k\|_\infty < \infty$. Then, for any $0 < \delta \leq 1$, with probability at least $1 - \delta$,*

$$\|\widehat{g}_k - g_k\|_2 \leq \frac{\|f_k\|_\infty}{\sqrt{\delta N\rho_k}}. \tag{12}$$

The radius grows as $(N\rho_k)^{-1/2}$. Thus, low ESS weakens the reliability guarantee for the IS gradient, consistent with the motivating example in the introduction. We now combine this radius with Lemma 1 to obtain an ESS-dependent improvement guarantee.

Theorem 3 (ESS-dependent improvement for the IS estimator)**.** *Under the conditions of Lemma 2, suppose that J is L_k-smooth along the update segment for $L_k > 0$. For any $0 < \delta \leq 1$, define*

$$\varepsilon_k(\delta) := \frac{\|f_k\|_\infty}{\sqrt{\delta N\rho_k}}, \qquad \gamma_k := \frac{1}{L_k}\left(1 - \frac{\varepsilon_k(\delta)}{\|\widehat{g}_k\|_2}\right)_+, \tag{13}$$

with $\gamma_k = 0$ when $\widehat{g}_k = 0$. Then, with probability at least $1 - \delta$,

$$J(\theta_k + \gamma_k\widehat{g}_k) - J(\theta_k) \geq \frac{1}{2L_k}\left(\|\widehat{g}_k\|_2 - \varepsilon_k(\delta)\right)_+^2. \tag{14}$$

Theorem 3 makes the role of ESS explicit. Its lower bound is positive when

$$\|\widehat{g}_k\|_2 > \frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}}.$$

Thus, the IS update remains certifiably useful while effective support is sufficiently large relative to the observed gradient signal. Holding the other terms fixed, both the admissible step and the certified improvement shrink as ESS falls. This complements population policy-deviation guarantees, which characterize the consequences of changing the policy, by controlling the finite-rollout direction used to make that change. Having established when the IS update is reliable, we next ask when an alternative gradient estimator provides the stronger one-step certificate.

4.2 Choosing an alternative gradient estimator

The preceding result connects the reliability of the IS estimator to policy improvement. When this guarantee weakens, one may instead use a controlled coefficient rule, such as truncated IS or PPO-style masking. We compare a general rule u with IS in two steps: first as a question of gradient estimation, and then as a question of policy improvement at a common fixed step. Write $\widehat{g}_k(u) := \widehat{g}_N(u;\theta_k)$. For a rule u, adding and subtracting the estimator mean gives the MSE decomposition

$$\begin{aligned} m_k(u) &:= \mathbb{E}\|\widehat{g}_k(u) - g_k\|_2^2 \\ &= \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2 + \mathbb{E}\|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2. \end{aligned} \tag{15}$$

The two terms are squared bias and sampling variation. Since the IS estimator is unbiased, a controlled rule has lower MSE exactly when its reduction in sampling variation exceeds the squared bias introduced by changing the coefficients. We call this the estimator-level bias–variance crossover. The next theorem gives the expected-improvement certificate used to compare such estimators at a common fixed step.

Theorem 4 (Improvement certificate)**.** *Suppose that J is L_k-smooth on every update segment generated by a coefficient rule u almost surely, for a constant $L_k > 0$, and that $\mathbb{E}\|\widehat{g}_k(u)\|_2^2 < \infty$. For any step size $\eta_k > 0$,*

$$\mathbb{E}\left[J(\theta_k + \eta_k \widehat{g}_k(u)) - J(\theta_k)\right] \geq \eta_k g_k^\top \mathbb{E}[\widehat{g}_k(u)] - \frac{L_k \eta_k^2}{2} \mathbb{E}\|\widehat{g}_k(u)\|_2^2. \tag{16}$$

Denote the right-hand side by

$$B_k(u;\eta_k) := \eta_k g_k^\top \mathbb{E}[\widehat{g}_k(u)] - \frac{L_k \eta_k^2}{2} \mathbb{E}\|\widehat{g}_k(u)\|_2^2. \tag{17}$$

The certificate has two parts. The first rewards alignment with the population gradient; the second penalizes the expected squared update under smoothness. Coefficient control improves the certificate only when the penalty it removes outweighs the useful alignment it removes. The next corollary states this crossover directly.

Corollary 5 (IS-to-controlled-estimator crossover)**.** *Let u_{IS} denote the IS coefficient rule and let u be any alternative rule. Suppose the same deterministic L_k is valid almost surely for both families of update segments. With a common step size $0 < \eta_k \leq 1/L_k$, the alternative rule has the stronger certificate if and only if*

$$B_k(u;\eta_k) > B_k(u_{\mathrm{IS}};\eta_k) \quad \Longleftrightarrow \quad m_k(u_{\mathrm{IS}}) - m_k(u) > (1 - L_k\eta_k)\left\{\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \tag{18}$$

When the alternative lowers the update second moment, the right-hand side of Equation (18) is its discounted reduction. Its MSE gain must exceed this amount. Since the IS estimator is unbiased, the same condition has the equivalent alignment form

$$\|g_k\|_2^2 - g_k^\top \mathbb{E}[\widehat{g}_k(u)] < \frac{L_k \eta_k}{2} \left\{ \mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2 \right\}. \tag{19}$$

The left-hand side is the population-gradient alignment lost by changing the estimator; the right-hand side is the reduction in the quadratic smoothness penalty.

Remark 6 (Truncated importance sampling). Upper truncation, $u_{\mathrm{T},t} = \min\{r_t, c\}$, caps each selected ratio without deleting its tail contribution. It is therefore a cheap variance safeguard with comparatively mild bias intervention. Its control is also limited: it acts only on the chosen upper tail, so the reduction in the update second moment can be modest under sequence-level mismatch.

Remark 7 (PPO-style masking). The PPO rule in Equation (6) removes saturated positive-advantage terms and low-ratio negative-advantage terms. It can reduce the update second moment more aggressively than upper truncation, but it also discards more population-gradient alignment. Equations (18) and (19) state when that stronger intervention is worthwhile.

5 Policy optimization and ESS

This section makes the theoretical distinction observable in two settings. An exactly solvable Gaussian bandit first separates gradient-estimation MSE from one-step policy improvement. Large-model trajectories then show how observed effective support evolves around delayed optimization failure.

5.1 Policy optimization in a Gaussian bandit

To isolate the estimator and optimization comparisons, we construct a scalar Gaussian bandit with a quadratic reward. Gaussian policies and smooth objectives provide a standard analytic setting for policy-gradient methods [5, 8, 10]. We hold the current policy fixed and vary only its distance from the rollout policy. Along this path, every quantity is available in closed form and contains no Monte Carlo error; Appendix A.1 gives the setup and calculations.

At each point on this path, the MSE oracle selects between IS and PPO by gradient-estimation MSE, whereas the certificate oracle selects the candidate with larger $B_k(u; \eta_k)$. The safe certificate oracle additionally includes a null update with certificate zero. We use $N = 32$, PPO radius $\epsilon = 0.2$, gradient $g = 2$, and step size $\eta = 0.4$, and vary ρ along one monotone ESS path.

Figure 2 first shows why MSE matters. As effective support contracts from $\rho = 0.961$ to 0.0109, the IS MSE rises from 0.228 to 62.3 and its exact expected improvement falls from 1.26 to -3.70. The same ESS path therefore reveals the reliability loss before choosing a particular intervention. It is a regime diagnostic, not the oracle: the pairwise IS–PPO MSE crossover occurs at $\rho_{\mathrm{MSE}} = 0.123$, while their certificate crossover occurs later at $\rho_B = 0.0392$.

The gap between the two crossovers shows why MSE alone does not determine the better update. PPO has lower MSE below ρ_{MSE}, but masking also removes population-gradient alignment. Its certificate therefore overtakes IS only below ρ_B, where the reduction in update variability finally compensates for the discarded signal. At still lower support, PPO reaches zero certificate at $\rho_{\mathrm{stop}} = 0.0109$ and the safe oracle abstains. These values are specific to (g, N, ϵ, η), but the ordering is the point: ESS locates the support regime, while MSE reduction and retained alignment jointly determine the better update.

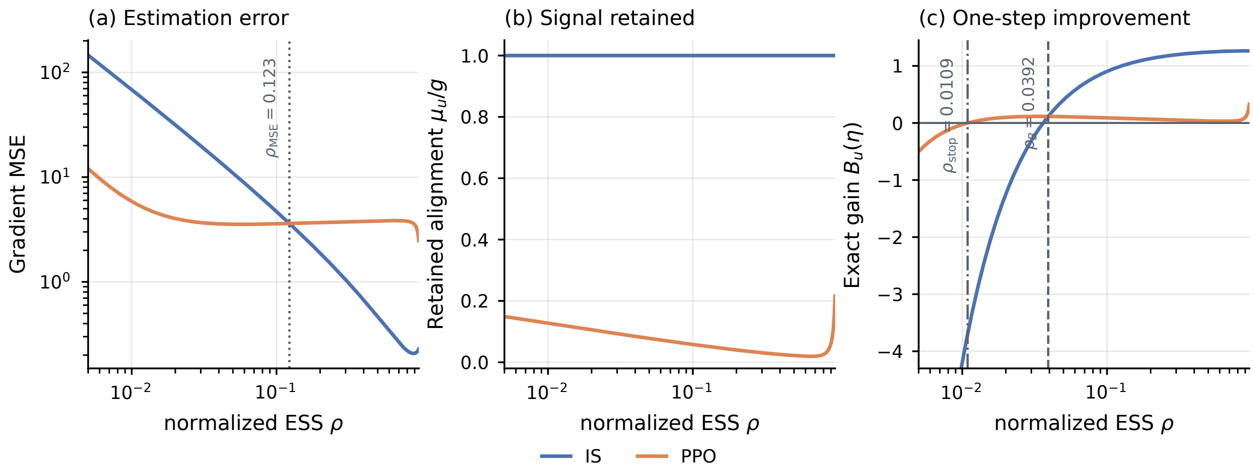

Figure 2: Exact Gaussian-bandit comparison with $g = 2$, $N = 32$, $\epsilon = 0.2$, and $\eta = 0.4$. (a) Gradient MSE as effective support contracts. (b) Population-gradient alignment retained by each estimator. (c) Exact expected one-step improvement, equal here to $B_u(\eta)$. The dotted line in panel (a) marks $\rho_{\mathrm{MSE}} = 0.123$; in panel (c), the dashed line marks $\rho_B = 0.0392$ and the dash-dotted line marks PPO's zero certificate at $\rho_{\mathrm{stop}} = 0.0109$.

5.2 Policy optimization in large language models

The Gaussian comparison holds the policy state fixed. Large-model training adds a temporal question: can observed effective support flag a sustained low-support regime before held-out performance begins to fall? We examine this ordering along an unclipped Qwen3-30B-A3B policy-gradient trajectory, evaluated on AIME-2024 using mean@16.

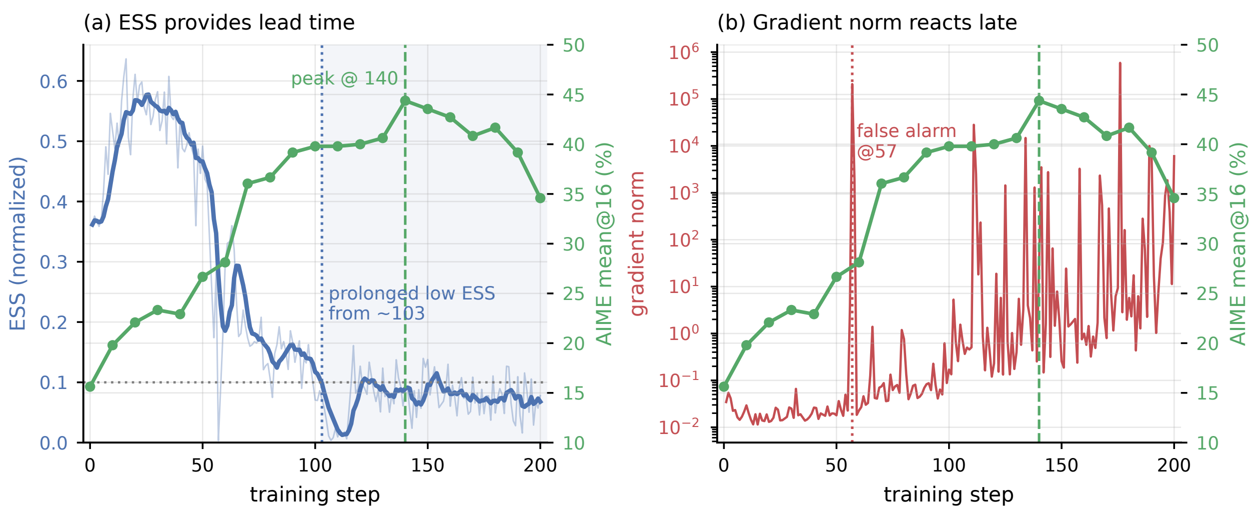

Figure 3: In this trajectory, normalized ESS flags low support before late performance degradation. (a) The trailing seven-update average of normalized ESS enters a prolonged sub-0.1 regime near step 103, whereas AIME-2024 mean@16 peaks at step 140 and subsequently declines. The thin curve is raw ESS. (b) Gradient norm spikes episodically: the large step-57 event is followed by continued improvement, while the largest event occurs only after the validation peak. Green curves report AIME-2024 mean@16.

ESS flags low support before performance deteriorates. Figure 3 shows the ordering directly. Smoothed ESS enters the prolonged low-support regime roughly 37 updates before AIME performance peaks at 44.4%; performance then falls by 9.79 percentage points to 34.6% at step 200. The diagnostic value lies in this lead time: support contraction becomes visible while optimization still appears successful, leaving a window in which to activate a safeguard. Gradient norm provides no comparable ordering. Its step-57 spike is followed by a 16.3-point improvement through step 140, whereas its largest spike at step 176 occurs after performance has peaked. Along this trajectory, ESS therefore provides the earlier support warning, while gradient norm mainly records episodic or already-emerging instability.

Together, the two examples separate diagnosis from action. The Gaussian bandit shows why effective support governs estimator reliability, and the large-model trajectory shows that sample ESS can expose support contraction before held-out degradation. ESS does not determine the low-support update; the next section tests whether using it to time a prespecified safeguard preserves permissive learning while preventing late failure.

6 ESS-conditioned policy optimization

Section 5 identifies ESS as an observable coordinate for effective support. We now ask whether acting on this signal resolves the central optimization tradeoff: retain a permissive update while the rollout remains informative, then activate control as effective support contracts. We first test this rule in Qwen3-30B-A3B math RLVR and then change the model scale and training objective.

6.1 Update rule and protocol

Let $\widehat{g}_{\text{perm}}$ denote the permissive update and $\widehat{g}_{\text{safe}}$ a prespecified safeguard. Given a threshold τ, the ESS-conditioned update is

$$\widehat{g}_{\text{gate}} = \begin{cases} \widehat{g}_{\text{perm}}, & \widehat{\rho} \geq \tau, \\ \widehat{g}_{\text{safe}}, & \widehat{\rho} < \tau. \end{cases} \tag{20}$$

ESS determines when the update changes; the selected safeguard determines how the low-support update is controlled. We fix $\tau = 0.1$ for all primary comparisons and vary it only in the threshold ablation. This value was prescribed for the language-model runs and is not derived from the Gaussian thresholds in Section 5.1.

Evaluation protocol. The main experiment trains Qwen3-30B-A3B for 200 policy updates on the boxed mathematical-reasoning data of Yu et al. [18] and evaluates AIME-2024 using mean@16. The primary permissive branch is TIS with ratio cap 3; an explicit ablation instead uses the unclipped IS update. At low ESS, we test three prespecified safeguards: conventional GRPO clipping [15], Clip-Higher [18], and DPPO [11].

6.2 Main results

ESS conditioning preserves learning while preventing collapse. Figure 4 establishes the central empirical result. Each conditional trajectory uses TIS 3 while effective support is high and activates the named safeguard only when normalized ESS falls below 0.1. Across GRPO, Clip-Higher, and DPPO, this construction achieves both a higher peak and a higher final AIME score than applying the same safeguard throughout training, with endpoint gains ranging from 6.88 to 12.3 percentage points. The direct TIS-3 comparison isolates the failure that conditioning prevents: ungated TIS 3 peaks at 31.7% and then collapses to 0.417%, whereas the ESS-conditioned trajectory reaches 47.3% and retains 44.8% at step 200. ESS conditioning therefore does more than stabilize a permissive estimator. It preserves its learning advantage while support is broad and introduces control only after effective support contracts.

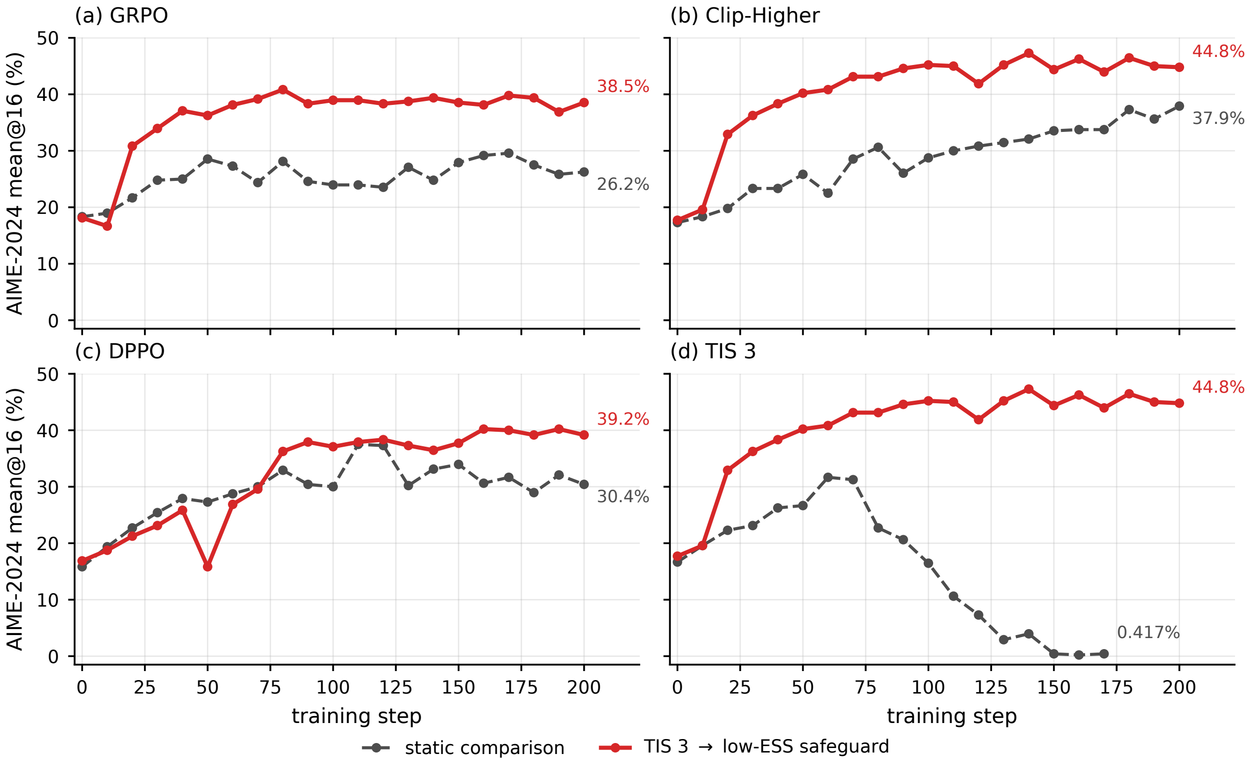

Figure 4: ESS conditioning combines permissive high-support learning with low-support control on Qwen3-30B-A3B. Each conditional trajectory uses TIS 3 when $\widehat{\rho} \geq 0.1$ and the named safeguard otherwise. Panels (a)–(c) compare this construction with applying GRPO, Clip-Higher, or DPPO throughout training. Panel (d) isolates collapse prevention against ungated TIS 3 using the TIS-3/Clip-Higher trajectory from panel (b). Direct labels report the final observed AIME-2024 mean@16.

ESS controls when intervention is needed. The realized intervention schedule confirms that conditioning changes the timing, rather than the definition, of each safeguard. For the DPPO construction, mean normalized ESS decreases from 0.412 over steps 1–50 to 0.120 over steps 151–200, while the safeguarded-update fraction increases from 0.153 to 0.668. Figure 10 shows the same ordering for all three safeguards. The gate therefore leaves the permissive estimator largely available during higher-support learning and strengthens protection as effective support contracts.

The timing principle extends beyond TIS. Figure 5 uses the unclipped trajectory diagnosed in Figure 3 as the permissive reference. Conditional clipping preserves a comparable peak and raises step-200 performance from 34.6% to 38.1%. Thus, the same ESS signal can retain most of the permissive estimator's learning while limiting late-stage degradation without relying on a particular truncation cap.

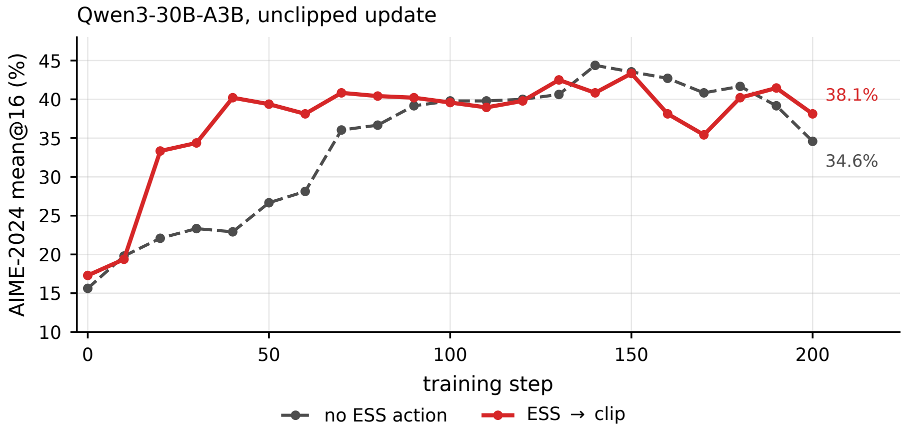

Figure 5: The timing principle extends to a fully unclipped high-support update. Conventional clipping is activated only when raw normalized ESS is below 0.1. Conditional control preserves a comparable peak while reducing the late performance loss. Direct labels report step-200 AIME-2024 mean@16.

The endpoint ordering remains at smaller scale. Figure 6 changes from the 30B mixture-of-experts model to dense Qwen3-8B while keeping math RLVR fixed. The ESS-conditioned construction finishes above its static counterpart in both aggregation-matched comparisons. The gains are smaller than the 30B rescue, but the endpoint ordering persists under both Clip-Higher and DPPO. This extends the support-conditioned design beyond the model on which the failure was first observed.

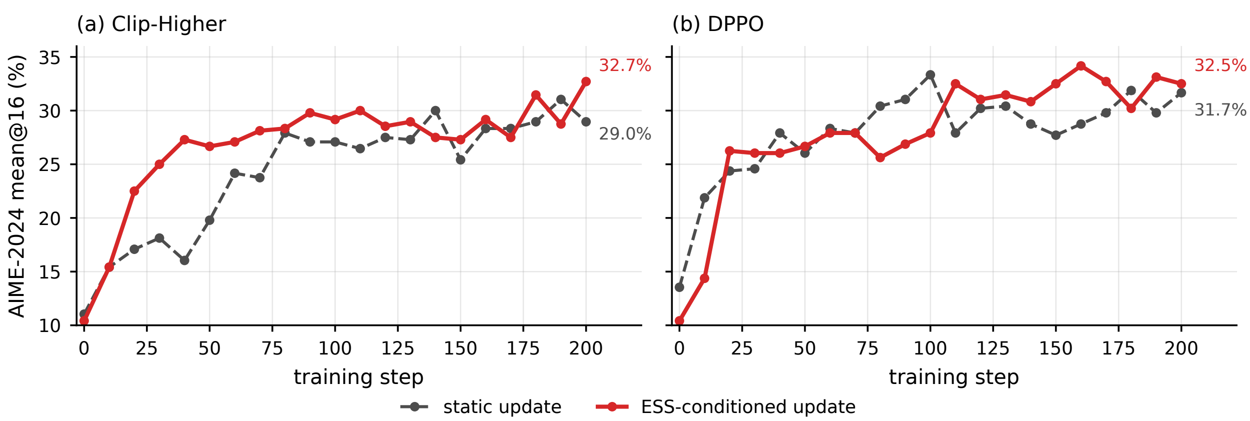

Figure 6: Dense-model transfer on Qwen3-8B. Panel (a) compares token-mean Clip-Higher with its token-mean ESS-conditioned construction. Panel (b) compares static and ESS-conditioned DPPO under the same sum-normalized aggregation. Direct labels give the step-200 AIME-2024 mean@16.

The same ordering appears under learned-reward optimization. We next change both the objective and the data, training Qwen3-4B-Instruct-2507 for 200 updates on Anthropic HH-RLHF [1] using Skywork-Reward-Llama-3.1-8B [4] as the reward model. Figure 7 shows that the complete TIS-3-plus-ESS recipe reaches a reward-model score of 60.7, compared with 43.0 for GRPO. The advantage therefore survives a shift from verifiable math rewards to learned-reward optimization.

This experiment measures optimization of the training reward model rather than human preference directly.

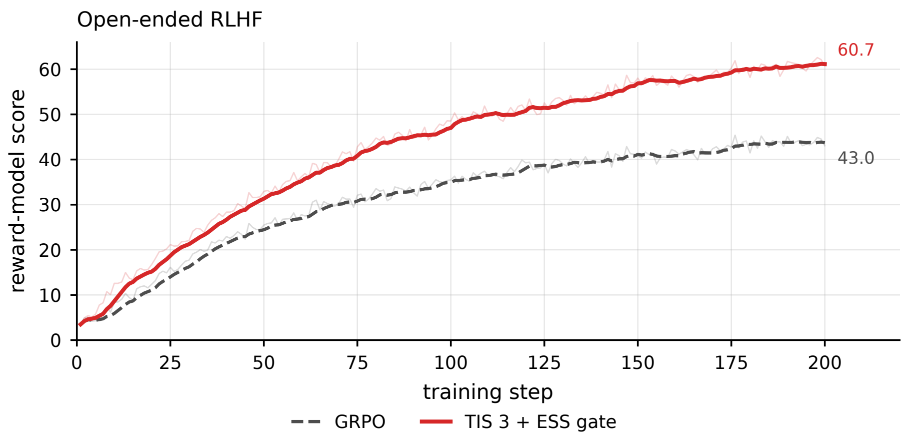

Figure 7: Transfer to open-ended RLHF on Qwen3-4B-Instruct-2507. Thin curves are per-step reward-model scores; thick curves are trailing seven-step means. Direct labels give the step-200 observations. This is a complete-recipe comparison between GRPO and TIS 3 with an ESS-conditioned safeguard.

Across these experiments, ESS contributes timing rather than a new clipping rule: it converts conservative control from a permanent constraint into a targeted intervention.

6.3 Ablations and alternative interventions

Collapse prevention does not depend on one exact threshold. Figure 8 varies only the ESS threshold for the TIS-3/GRPO construction. Thresholds 0.05, 0.1, and 0.2 all avoid the catastrophic 0.417% endpoint of ungated TIS 3, with last-reported scores of 43.1%, 38.5%, and 34.8%, respectively. Performance changes because the threshold determines how long the permissive branch remains active. Thus, τ is an operating point for the ESS diagnostic rather than a universal theoretical constant; the failure-prevention effect persists across the tested fourfold range.

Alternative low-support actions separate detection from control. ESS can also trigger a complete update veto rather than clipping. Figure 9 shows that this simpler action rescues TIS 5 from a 0.00% collapse to a 39.4% endpoint. Skipping is more conservative when the permissive trajectory remains viable: with the unclipped update it finishes at 35.6%, below the 38.1% obtained by ESS-conditioned clipping, and sacrifices more of the attainable peak. Across both experiments, ESS provides a common trigger for the low-support regime, but the response still matters. The stronger performance under conditional clipping is consistent with controlled estimation retaining useful signal that a full update veto discards.

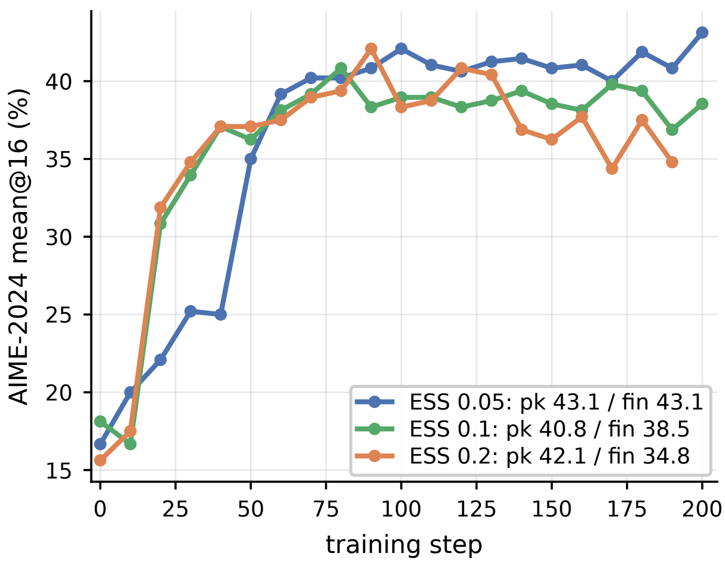

Figure 8: Sensitivity to the ESS threshold in the Qwen3-30B-A3B TIS-3/GRPO construction. All three thresholds prevent the ungated TIS-3 collapse; changing τ shifts the balance between the permissive and safeguarded branches. Labels report peak and last-observed AIME-2024 mean@16.

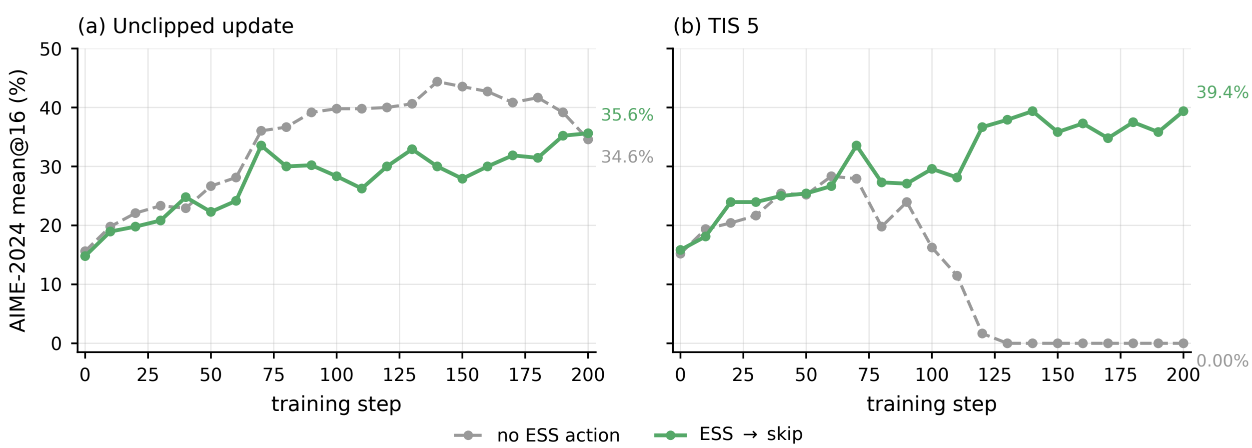

Figure 9: ESS-conditioned skipping as an alternative low-support action. Panel (a) shows that a hard veto stabilizes an unclipped update but sacrifices more learning than ESS-conditioned clipping. Panel (b) gives the matched TIS-5 comparison, in which skipping prevents the static trajectory's collapse. All curves report AIME-2024 mean@16.

Together, the threshold and action ablations show that collapse prevention comes from conditioning control on effective support, while retained learning depends on how aggressively the low-support update is modified.

7 Conclusion

We studied clipping as a finite-sample estimator choice rather than a default property of policy optimization. The theory separates gradient-estimation error from the useful update retained after control: reducing MSE improves the fixed-step certificate only when that gain compensates for the signal removed by clipping. This distinction makes IS preferable while effective support is broad and explains why controlled estimators become attractive as mismatch grows. The exact Gaussian construction exhibits separate MSE and update crossovers. The reported language-model trajectories then show that sequence ESS can reveal support contraction and time a prespecified safeguard without serving as the oracle itself. In the RLVR and RLHF examples, the resulting conditional constructions retain permissive learning in the higher-support regime and protect training more strongly as support contracts. ESS therefore provides the operational link between importance-weight reliability and the decision of when to control an update.

A Numerical Experiments

A.1 Gaussian bandit

This section gives the calculations used in Section 5.1.

Setup We compare IS and the PPO-style rule in Equation (6), using the exact current-policy value baseline.

Because the score contribution is polynomial and each mask boundary is linear in z, the means, MSEs, and update second moments are finite sums of truncated Gaussian moments.

We choose a quadratic reward because it makes the smoothness certificate equal to the true expected gain. Any separation between the two oracles is therefore genuine rather than an artifact of a loose bound.

Consider the current policy and rollout policy

$$P_\theta = \mathcal{N}(\theta, 1), \qquad Q_\Delta = \mathcal{N}(\theta - \Delta, 1), \qquad \Delta \geq 0, \tag{21}$$

with reward $R(a) = -\frac{1}{2}(a - a^\star)^2$. We hold the current policy fixed and vary only the rollout gap Δ. Writing $g := a^\star - \theta > 0$, the objective is

$$J(\theta) = -\frac{1}{2}\{(\theta - a^\star)^2 + 1\}, \qquad \nabla J(\theta) = g, \qquad L = 1. \tag{22}$$

Here θ is the fixed pre-update mean and Q_Δ remains frozen during the hypothetical one-step update. We vary Δ only across common-state comparisons. For $z = a - \theta$, the likelihood ratio and normalized ESS reduce to

$$W_\Delta(z) = \exp\{\Delta z + \Delta^2/2\}, \qquad \rho(\Delta) = \{\mathbb{E}_{Q_\Delta}[W_\Delta^2]\}^{-1} = \exp(-\Delta^2). \tag{23}$$

Normalized ESS is therefore a one-to-one coordinate for this branch.

Write $\mu_u := \mathbb{E}[\widehat{g}_u]$, $s_u := \mathbb{E}[\widehat{g}_u^2]$, and $B_u(\eta) := \eta g \mu_u - \eta^2 s_u/2$. For this quadratic objective, the fixed-step certificate is exact:

$$\mathbb{E}[J(\theta + \eta\widehat{g}_u) - J(\theta)] = \eta g \mu_u - \frac{\eta^2}{2} s_u = B_u(\eta). \tag{24}$$

We use $N = 32$, the standard PPO radius $\epsilon = 0.2$, and choose $g = 2$ and $\eta = 0.4$ so that the MSE, improvement, and abstention regimes are all visible on one monotone ESS path.

Exact Closed Form Computations Let $z = a - \theta$. Under the current policy, $z \sim \mathcal{N}(0, 1)$ and the score is z. The exact current-policy value baseline gives

$$A_g(z) = gz + \frac{1 - z^2}{2}, \qquad f_g(z) := zA_g(z) = gz^2 + \frac{z - z^3}{2}. \tag{25}$$

Although f_g is unbounded, all polynomial Gaussian moments used below are finite. The example therefore uses the finite-moment fixed-step certificate; the bounded-integrand IS theorem addresses a separate reliability question. IS uses $W_\Delta f_g$. The PPO-style candidate uses $W_\Delta f_g M_{\epsilon,\Delta}$, where

$$M_{\epsilon,\Delta}(z) := \mathbf{1}\{A_g(z) \geq 0,\ W_\Delta(z) \leq 1 + \epsilon\} + \mathbf{1}\{A_g(z) < 0,\ W_\Delta(z) \geq 1 - \epsilon\}. \tag{26}$$

Let $M_{\mathrm{IS}} \equiv 1$ and $M_{\mathrm{P}} = M_{\epsilon,\Delta}$. Change of measure gives the exact per-sample moments

$$\begin{aligned} \mu_u(\Delta) &= \mathbb{E}_{U\sim\mathcal{N}(0,1)}[f_g(U)M_u(U)], \\ h_u(\Delta) &= e^{\Delta^2}\mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[f_g(V)^2 M_u(V)], \qquad u \in \{\mathrm{IS}, \mathrm{P}\}. \end{aligned} \tag{27}$$

In particular, $\mu_{\mathrm{IS}} = g$. For an iid estimator with batch size N,

$$m_u = (\mu_u - g)^2 + \frac{h_u - \mu_u^2}{N}, \qquad s_u = \mu_u^2 + \frac{h_u - \mu_u^2}{N}. \tag{28}$$

The advantage changes sign at $g \pm \sqrt{g^2 + 1}$, and the ratio boundaries are $(\log(1 \pm \epsilon) - \Delta^2/2)/\Delta$. Equation (27) is therefore a finite sum of truncated normal polynomial moments. With $I_j(a, b) := \int_a^b z^j \phi(z)\, dz$, integration by parts gives

$$I_j(a, b) = a^{j-1}\phi(a) - b^{j-1}\phi(b) + (j - 1)I_{j-2}(a, b), \qquad j \geq 2, \tag{29}$$

starting from $I_0 = \Phi(b) - \Phi(a)$ and $I_1 = \phi(a) - \phi(b)$. Boundary terms vanish at infinity. Substituting these moments into Equation (28) and the exact gain in Equation (24) gives the reported boundaries.

Results We evaluate $0.2 \leq \Delta \leq \sqrt{-\log(0.005)}$. Numerical evaluation of the closed-form moments gives three substantive boundaries on this branch:

$$\rho_{\mathrm{MSE}} = 0.123, \qquad \rho_B = 0.0392, \qquad \rho_{\mathrm{stop}} = 0.0109. \tag{30}$$

The MSE oracle over $\mathcal{U} = \{u_{\mathrm{IS}}, u_{\mathrm{P}}\}$ selects IS above ρ_{MSE} and PPO below it. The safe certificate oracle adds the null update and instead selects IS for $\rho > \rho_B$, PPO for $\rho_{\mathrm{stop}} < \rho < \rho_B$, and no update for $\rho < \rho_{\mathrm{stop}}$. Adjacent choices tie at each boundary.

B Additional coefficient-rule details

The main text treats coefficient control generically because different rules alter different parts of the gradient. A factorwise detached upper truncation uses $u_{\mathrm{T},t}(\mathbf{r}, A) := \min\{r_t, c\}$. Its contribution obeys

$$\|\psi_k(u_{\mathrm{T}}; Z)\|_2 \le c|A(Z)| \sum_{t=1}^{T} \|\ell_{\theta_k,t}(Z)\|_2, \tag{31}$$

so it has a direct second-moment envelope whenever the right-hand side has a finite second moment.

PPO masking is different. For $A \ge 0$ and $c = 1 + \epsilon$,

$$0 \le r - \min\{r, c\} = (r - c)_+ \le r\mathbf{1}\{r > c\} = r - rM_\epsilon(r, A), \tag{32}$$

where $M_\epsilon(r, A) = \mathbf{1}\{A \ge 0,\ r \le 1+\epsilon\} + \mathbf{1}\{A < 0,\ r \ge 1-\epsilon\}$. Truncation retains a bounded positive coefficient above the cap, whereas PPO removes that contribution. On the negative-advantage lower branch, PPO masks small ratios whereas upper truncation does not. A complete-sample mask is another distinct rule: it sets $u_t(\mathbf{r}, A) = W_k M_\epsilon(W_k, A)$ for every factor. We state which rule is used in each experiment rather than treating these implementations as interchangeable.

C Additional RLVR diagnostics

C.1 Intervention and training dynamics

The main text reports evaluation performance. We record here how often the conditional rule intervenes and the accompanying training dynamics.

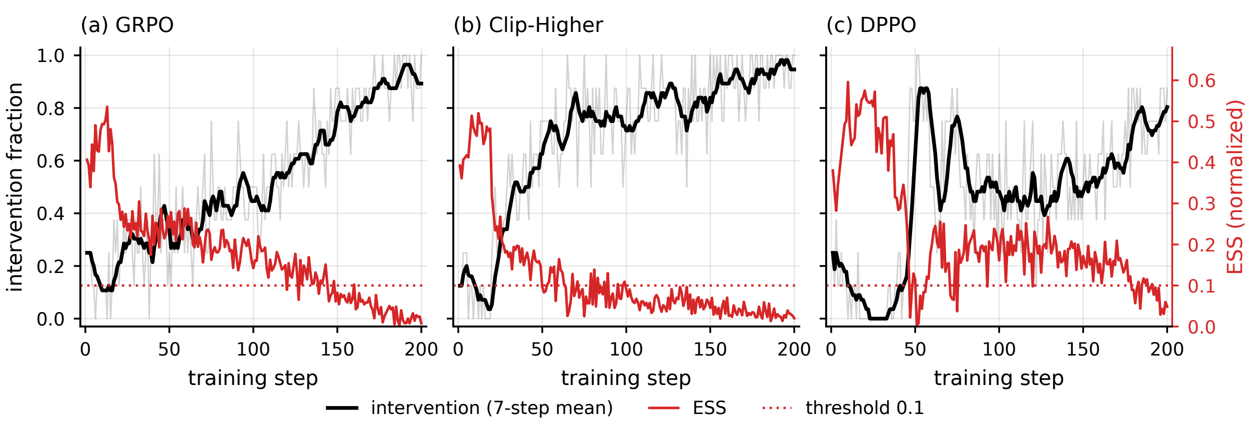

Figure 10: Realized ESS-conditioned intervention schedules. Each panel pairs the per-step intervention fraction (thin black) and its trailing seven-step mean (thick black) with normalized ESS (red). GRPO and DPPO use raw ESS; Clip-Higher uses shaped ESS after the coefficient cap. The dotted line marks the threshold 0.1.

Figure 10 shows the intended temporal ordering: intervention is already possible early, but its frequency increases as ESS contracts. The remaining figures record response length, entropy, and reward diagnostics.

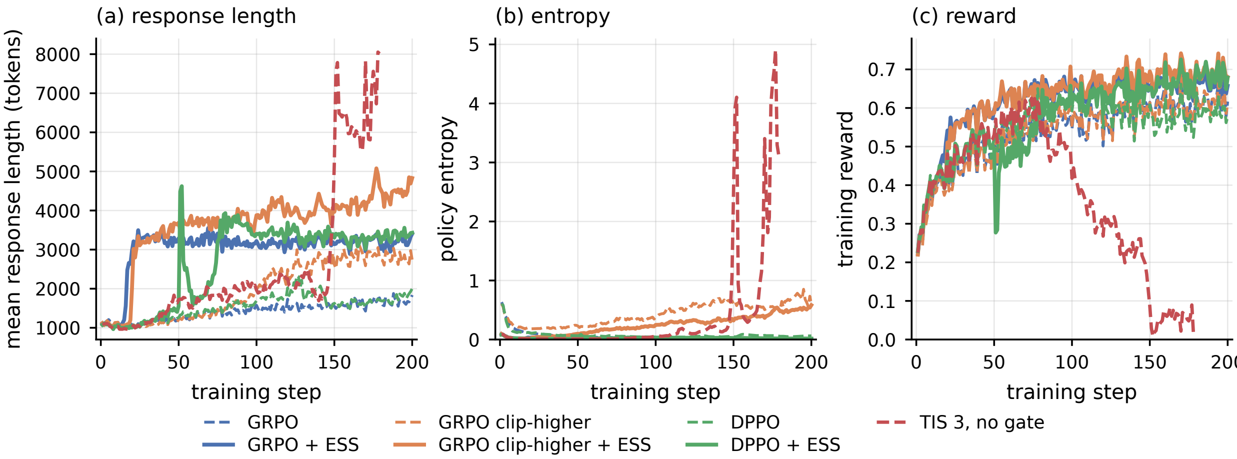

Figure 11: Training dynamics for the main comparisons. The static TIS 3 run develops extreme response length and entropy together with a collapse in training reward. The safeguards and their ESS-conditional variants avoid this failure over the reported 200 steps.

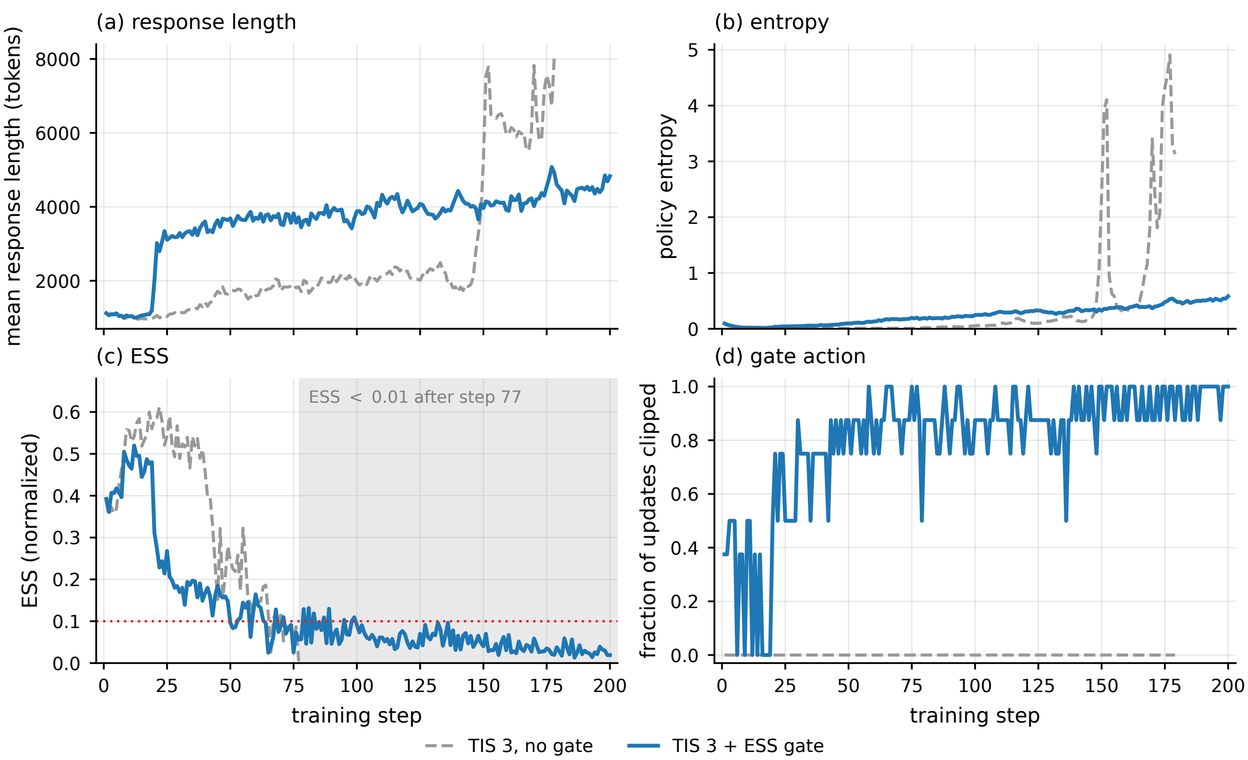

Figure 12: Detailed TIS 3 comparison. Under static TIS, ESS falls below 0.01 after step 77 and the response length and entropy diverge. The fixed-0.1 intervention increases the clipped fraction as ESS declines and keeps both diagnostics controlled.

D Proofs for the theoretical results

At update k, all expectations in this appendix are taken after conditioning on the preceding updates. The current policy is therefore fixed and the unused sampling units are independent draws from Q. We suppress this conditioning throughout.

D.1 Policy-gradient identity and mean-square bound

By definition, $f_k = (R - b(X))\nabla_\theta \log P_{\theta_k}(Y \mid X)$ and $W_k = dP_{\theta_k}/dQ$. Change of measure gives

$$\begin{aligned}\mathbb{E}_Q[W_k f_k] &= \mathbb{E}_{P_{\theta_k}}[(R - b(X))\nabla_\theta \log P_{\theta_k}(Y \mid X)] \\ &= \mathbb{E}_{P_{\theta_k}}[R\nabla_\theta \log P_{\theta_k}(Y \mid X)] \\ &\quad - \mathbb{E}_{P_{\theta_k}}[b(X)\nabla_\theta \log P_{\theta_k}(Y \mid X)]. \end{aligned} \tag{33}$$

For the baseline term, condition further on X. Under the standard regularity condition that permits differentiation under the conditional integral,

$$\begin{aligned}&\mathbb{E}_{P_{\theta_k}}[b(X)\nabla_\theta \log P_{\theta_k}(Y \mid X) \mid X] \\ &\quad = b(X) \int P_{\theta_k}(y \mid X)\nabla_\theta \log P_{\theta_k}(y \mid X)\, d\lambda(y) \\ &\quad = b(X)\nabla_\theta \int P_{\theta_k}(y \mid X)\, d\lambda(y) \\ &\quad = b(X)\nabla_\theta 1 = 0. \end{aligned} \tag{34}$$

The remaining term is the score-function identity. Hence

$$\mathbb{E}_Q[W_k f_k] = \nabla J(\theta_k) = g_k. \tag{35}$$

Averaging N independent copies gives $\mathbb{E}[\widehat{g}_k] = g_k$.

For the mean-square calculation, define $\xi_{k,i} := W_{k,i} f_{k,i} - g_k$. Equation (35) implies $\mathbb{E}[\xi_{k,i}] = 0$. Since $\widehat{g}_k - g_k = N^{-1}\sum_{i=1}^N \xi_{k,i}$,

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N^2}\sum_{i=1}^N \mathbb{E}\|\xi_{k,i}\|_2^2 + \frac{1}{N^2}\sum_{i\neq j}\mathbb{E}[\xi_{k,i}^\top \xi_{k,j}]. \tag{36}$$

For $i \neq j$, independence and zero means give

$$\mathbb{E}[\xi_{k,i}^\top \xi_{k,j}] = \mathbb{E}[\xi_{k,i}]^\top \mathbb{E}[\xi_{k,j}] = 0. \tag{37}$$

The diagonal terms are identical, so

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\mathbb{E}_Q\|W_k f_k - g_k\|_2^2. \tag{38}$$

Expanding the remaining squared norm yields

$$\begin{aligned}\mathbb{E}_Q\|W_k f_k - g_k\|_2^2 &= \mathbb{E}_Q[W_k^2\|f_k\|_2^2] - 2g_k^\top \mathbb{E}_Q[W_k f_k] + \|g_k\|_2^2 \\ &= \mathbb{E}_Q[W_k^2\|f_k\|_2^2] - \|g_k\|_2^2, \end{aligned} \tag{39}$$

where the last equality uses Equation (35). Combining Equations (38) and (39) gives

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\left\{\mathbb{E}_Q[W_k^2\|f_k\|_2^2] - \|g_k\|_2^2\right\}. \tag{40}$$

Finally, $\|f_k(z)\|_2 \le \|f_k\|_\infty$ and $\mathbb{E}_Q[W_k^2] = \rho_k^{-1}$ imply

$$\begin{aligned}\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 &\le \frac{1}{N}\mathbb{E}_Q[W_k^2\|f_k\|_2^2] \\ &\le \frac{\|f_k\|_\infty^2}{N}\mathbb{E}_Q[W_k^2] \\ &= \frac{\|f_k\|_\infty^2}{N\rho_k},\end{aligned} \tag{41}$$

which proves Equation (11).

D.2 Proof of Lemma 2

Apply Markov's inequality to the nonnegative random variable $\|\widehat{g}_k - g_k\|_2^2$:

$$\begin{aligned}&\Pr\left(\|\widehat{g}_k - g_k\|_2 > \frac{\|f_k\|_\infty}{\sqrt{\delta N\rho_k}}\right) \\ &= \Pr\left(\|\widehat{g}_k - g_k\|_2^2 > \frac{\|f_k\|_\infty^2}{\delta N\rho_k}\right) \\ &\le \frac{\mathbb{E}\|\widehat{g}_k - g_k\|_2^2}{\|f_k\|_\infty^2/(\delta N\rho_k)} \\ &\le \delta,\end{aligned} \tag{42}$$

where the last line uses Equation (11). Taking complements proves Equation (12).

D.3 Proof of Lemma 1

Let $\|\widehat{g} - g\|_2 \le \varepsilon$. Smoothness gives, for every $\gamma \ge 0$,

$$J(\theta + \gamma\widehat{g}) - J(\theta) \ge \gamma g^\top \widehat{g} - \frac{L\gamma^2}{2}\|\widehat{g}\|_2^2. \tag{43}$$

The inner product satisfies

$$\begin{aligned}g^\top\widehat{g} &= \|\widehat{g}\|_2^2 + (g - \widehat{g})^\top\widehat{g} \\ &\ge \|\widehat{g}\|_2^2 - \|g - \widehat{g}\|_2\|\widehat{g}\|_2 \\ &\ge \|\widehat{g}\|_2(\|\widehat{g}\|_2 - \varepsilon).\end{aligned} \tag{44}$$

Substituting Equation (44) into Equation (43) gives

$$J(\theta + \gamma\widehat{g}) - J(\theta) \ge \gamma\|\widehat{g}\|_2(\|\widehat{g}\|_2 - \varepsilon) - \frac{L\gamma^2}{2}\|\widehat{g}\|_2^2. \tag{45}$$

For $\widehat{g} \ne 0$, the derivative of the right-hand side is

$$\|\widehat{g}\|_2(\|\widehat{g}\|_2 - \varepsilon) - L\gamma\|\widehat{g}\|_2^2. \tag{46}$$

Its nonnegative maximizer is $\gamma = L^{-1}(1 - \varepsilon/\|\widehat{g}\|_2)_+$. When $\widehat{g} = 0$, both the chosen step and the claimed lower bound are zero. Substituting the maximizer into Equation (45) proves Equation (9).

D.4 Proof of Theorem 3

Lemma 2 shows that the event

$$\|\widehat{g}_k - g_k\|_2 \le \frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}} \tag{47}$$

has probability at least $1-\delta$. On this event, apply Lemma 1 with $\theta = \theta_k$, $\widehat{g} = \widehat{g}_k$, $\varepsilon = \|f_k\|_\infty/\sqrt{\delta N \rho_k}$, and $L = L_k$. The resulting inequality is exactly Equation (14).

D.5 Risk and certificate comparisons for coefficient rules

For a coefficient rule u, add and subtract the estimator mean:

$$\widehat{g}_k(u) - g_k = \{\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\} + \{\mathbb{E}[\widehat{g}_k(u)] - g_k\}. \tag{48}$$

Squaring and taking expectation gives

$$\begin{aligned} m_k(u) &= \mathbb{E}\,\|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2 + \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2 \\ &\quad + 2\{\mathbb{E}[\widehat{g}_k(u)] - g_k\}^\top \mathbb{E}\left[\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\right] \\ &= \mathbb{E}\,\|\widehat{g}_k(u) - \mathbb{E}[\widehat{g}_k(u)]\|_2^2 + \|\mathbb{E}[\widehat{g}_k(u)] - g_k\|_2^2, \end{aligned} \tag{49}$$

because the centered term has mean zero. This proves Equation (15). For an iid sample mean, the first term in Equation (49) is the per-sample variation divided by N.

It remains to connect total risk to the expected policy-improvement certificate. For any random estimate $\widehat{g}$ of g_k, smoothness gives

$$J(\theta_k + \eta_k \widehat{g}) - J(\theta_k) \ge \eta_k g_k^\top \widehat{g} - \frac{L_k \eta_k^2}{2}\|\widehat{g}\|_2^2. \tag{50}$$

Taking expectation in Equation (50), with $\mathbb{E}\|\widehat{g}\|_2^2 < \infty$, gives

$$\mathbb{E}[J(\theta_k + \eta_k \widehat{g}) - J(\theta_k)] \ge \eta_k g_k^\top \mathbb{E}[\widehat{g}] - \frac{L_k \eta_k^2}{2}\mathbb{E}\|\widehat{g}\|_2^2. \tag{51}$$

Setting $\widehat{g} = \widehat{g}_k(u)$ proves Theorem 4. The polarization identity, followed by expectation, gives

$$2 g_k^\top \mathbb{E}[\widehat{g}] = \|g_k\|_2^2 + \mathbb{E}\|\widehat{g}\|_2^2 - \mathbb{E}\|\widehat{g} - g_k\|_2^2. \tag{52}$$

Substituting Equation (52) into the certificate and subtracting the IS certificate gives

$$\begin{aligned} B_k(u;\eta_k) - B_k(u_{\mathrm{IS}};\eta_k) &= \frac{\eta_k}{2}\left[m_k(u_{\mathrm{IS}}) - m_k(u)\right] \\ &\quad - \frac{\eta_k}{2}(1 - L_k\eta_k)\left\{\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \end{aligned} \tag{53}$$

Since $\eta_k > 0$, the right-hand side is positive exactly under Equation (18), proving Corollary 5. Since the IS estimator is unbiased, subtracting the certificate definitions instead gives

$$\begin{aligned} B_k(u;\eta_k) - B_k(u_{\mathrm{IS}};\eta_k) &= -\eta_k\left\{\|g_k\|_2^2 - g_k^\top \mathbb{E}[\widehat{g}_k(u)]\right\} \\ &\quad + \frac{L_k\eta_k^2}{2}\left\{\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \end{aligned} \tag{54}$$

Its positivity is equivalent to Equation (19).

D.6 Intact prompt groups

Suppose M prompts are independent and each prompt has G conditionally independent responses. Keep each prompt group intact and define its contribution by $\Xi_{k,i} := G^{-1}\sum_{j=1}^{G} W_{k,ij} f_{k,ij}$. A leave-one-out baseline is admissible because it excludes the focal response. Conditional on the prompt and the remaining responses, the focal score still has mean zero. Hence $\mathbb{E}[\Xi_{k,i}] = g_k$.

Jensen's inequality gives

$$\begin{aligned}\|\Xi_{k,i}\|_2^2 &= \left\| \frac{1}{G}\sum_{j=1}^{G} W_{k,ij} f_{k,ij} \right\|_2^2 \\ &\le \frac{1}{G}\sum_{j=1}^{G} W_{k,ij}^2 \|f_{k,ij}\|_2^2. \end{aligned} \tag{55}$$

Repeating the sample-mean calculation across the M independent prompt groups gives

$$\begin{aligned}\mathbb{E}\left\| \frac{1}{M}\sum_{i=1}^{M} \Xi_{k,i} - g_k \right\|_2^2 &\le \frac{1}{M}\mathbb{E}\|\Xi_{k,1}\|_2^2 \\ &\le \frac{\|f_k\|_\infty^2}{M\rho_k}. \end{aligned} \tag{56}$$

Markov's inequality then gives the same form as Lemma 2 with the number of independent prompts M in place of the number of responses N.

D.7 Finite-moment relaxation

The exact equality in Equation (10) requires only $\mathbb{E}_Q[W_k^2\|f_k\|_2^2] < \infty$. The bounded-contribution condition is used solely to obtain the standard form $\|f_k\|_\infty^2/(N\rho_k)$. More generally, define $M_{2,k} := \mathbb{E}_Q[W_k^2\|f_k\|_2^2]/\mathbb{E}_Q[W_k^2]$. The same calculation gives $\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 \le M_{2,k}/(N\rho_k)$. We use the bounded form in the main text to match conventional importance-sampling notation and make the role of normalized ESS explicit.

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